Acoustic systems（3rd class）

- Acoustic characteristic impedance
- Sound propagation in 3D.
- The 1st exam

Review of velocity potential

(Previous) Question 1

Try to derive the wave equation,

$$\frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = \frac{\partial^2 p}{\partial x^2}$$

from the following equations.

Motion equation $\rho_0 \frac{\partial v}{\partial t} = -\frac{\partial p}{\partial x}$

Conservation of mass $\frac{\partial \rho}{\partial t} = -\rho_0 \frac{\partial v}{\partial x}$

Gas state equation $p = c^2 \rho \quad (where \quad c^2 = \frac{\gamma P_0}{\rho_0})$

(Previous) Q1 Answer

By taking the partial derivative of both sides of the motion equation with respect to x,

$$\frac{\partial}{\partial x}\left(\rho_0 \frac{\partial v}{\partial t}\right) = -\frac{\partial}{\partial x}\left(\frac{\partial p}{\partial x}\right) \rightarrow \rho_0 \frac{\partial^2 v}{\partial x \partial t} = \frac{\partial^2 p}{\partial x^2}$$

By substituting the law of mass conservation, $\rho_0 \frac{\partial^2 \rho}{\partial t^2} = \frac{\partial^2 p}{\partial x^2}$

By substituting the gas state equation, we can get the following wave equation finally.

$$\frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = \frac{\partial^2 p}{\partial x^2}$$

Important relation between ω , k and c

$\boldsymbol{k}$ is wave number. $k = \frac{2\pi}{\lambda}$ $\boldsymbol{c}$ is propagation speed.

ω is angular frequency. $\omega = 2\pi f$

Relation between ω , k and c is the following equation.

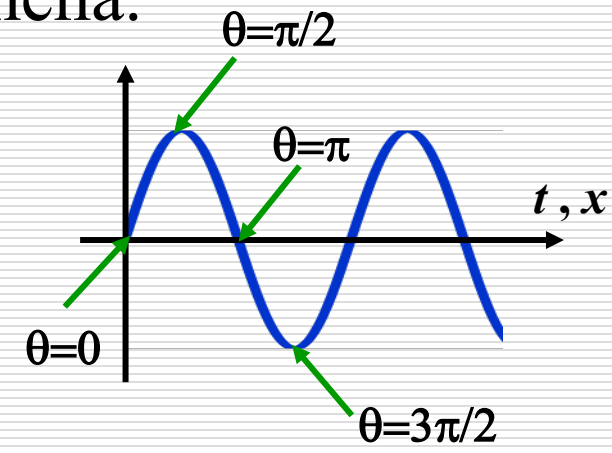
$$k = \frac{\omega}{c} \quad \Longleftrightarrow \quad c = f\lambda$$

same relation

Meanings of ω and k.

A phase is important in wave phenomena.

Acoustic phenomena only treats harmonic wave (sinusoidal wave).



k converts the distance x into the phase information θ.

$$\theta = kx$$

ω converts the time t into the phase information θ.

$$\theta = \omega t$$

Sound pressure, particle velocity, velocity potential

Sound pressure $\boldsymbol{p}$ can be obtained by the derivation of velocity potential $\boldsymbol{\Phi}$ in the time variable.

$$p = \rho \frac{\partial \Phi}{\partial t}$$

Density notation:
Instead of ρ_0, a static media density is expressed by ρ.

Particle velocity $\boldsymbol{v}$ can be obtained by the derivation of velocity potential $\boldsymbol{\Phi}$ in the spatial variable.

$$v = -\frac{\partial \Phi}{\partial x}$$

Relation between sound pressure and particle velocity

Wave equation of the sound pressure $\frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = \frac{\partial^2 p}{\partial x^2}$

Wave equation of the particle velocity $\frac{1}{c^2}\frac{\partial^2 v}{\partial t^2} = \frac{\partial^2 v}{\partial x^2}$

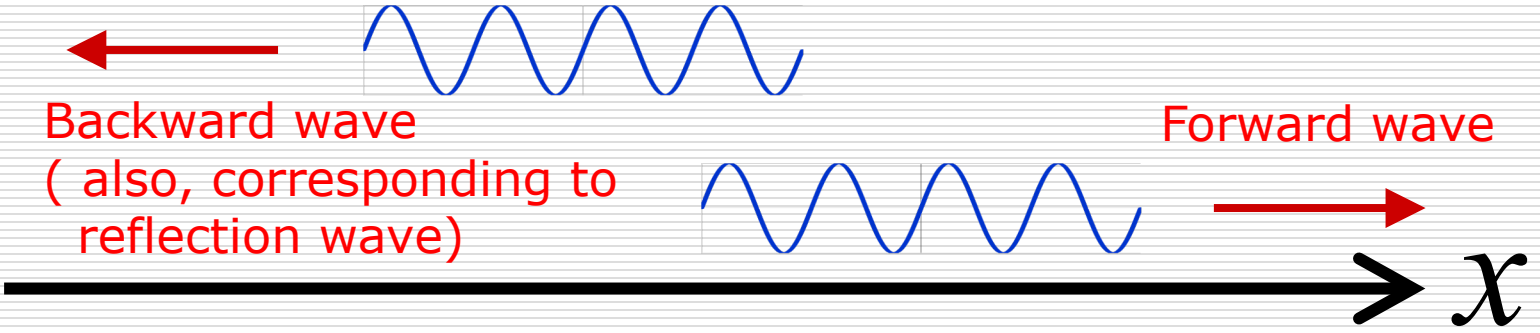
Velocity potential **Φ** is introduced in order to treat sound pressure and particle velocity simultaneously.

Wave equation of the velocity potential $\frac{1}{c^2}\frac{\partial^2 \Phi}{\partial t^2} = \frac{\partial^2 \Phi}{\partial x^2}$

Solution on wave equation of velocity potential

$$\Phi(x,t) = C_1 e^{j(\omega t - kx)} + C_2 e^{j(\omega t + kx)}$$

Forward wave　　Backward wave



Without reflection, only forward wave propagates.

$$\Phi(x,t) = C_1 e^{j(\omega t - kx)}$$

About time variation

In the acoustic phenomena, only harmonic wave (sinusoidal wave) is treated. Therefore, time variation term is always $C_1 e^{j\omega t}$.

The time variation term is often omitted.

(Previous) Question 2

The sound wave goes forward only, as written in the equation below. Find the sound pressure p and the particle velocity v, and calculate p/v.

$$\Phi(x,t) = C_1 e^{j(\omega t - kx)}$$

(Previous) Q2 Answer

$$p = \rho \frac{\partial \Phi}{\partial t} = \rho \frac{\partial \left(C_1 e^{j(\omega t - kx)} \right)}{\partial t} = j\omega\rho C_1 e^{j(\omega t - kx)} = j\omega\rho\Phi$$

$$v = -\frac{\partial \Phi}{\partial x} = -\frac{\partial \left(C_1 e^{j(\omega t - kx)} \right)}{\partial x} = jkC_1 e^{j(\omega t - kx)} = jk\Phi$$

$$\frac{p}{v} = \frac{j\omega\rho\Phi}{jk\Phi} = \frac{\omega\rho}{k} = \rho c$$

Acoustic characteristic impedance

Acoustic characteristic impedance

Ratio between sound pressure and particle velocity is called as acoustic characteristic impedance (or specific acoustic impedance)

Acoustic characteristic impedance

$$Z_s = \frac{p}{v} = \rho \frac{\omega}{k} = \rho c$$

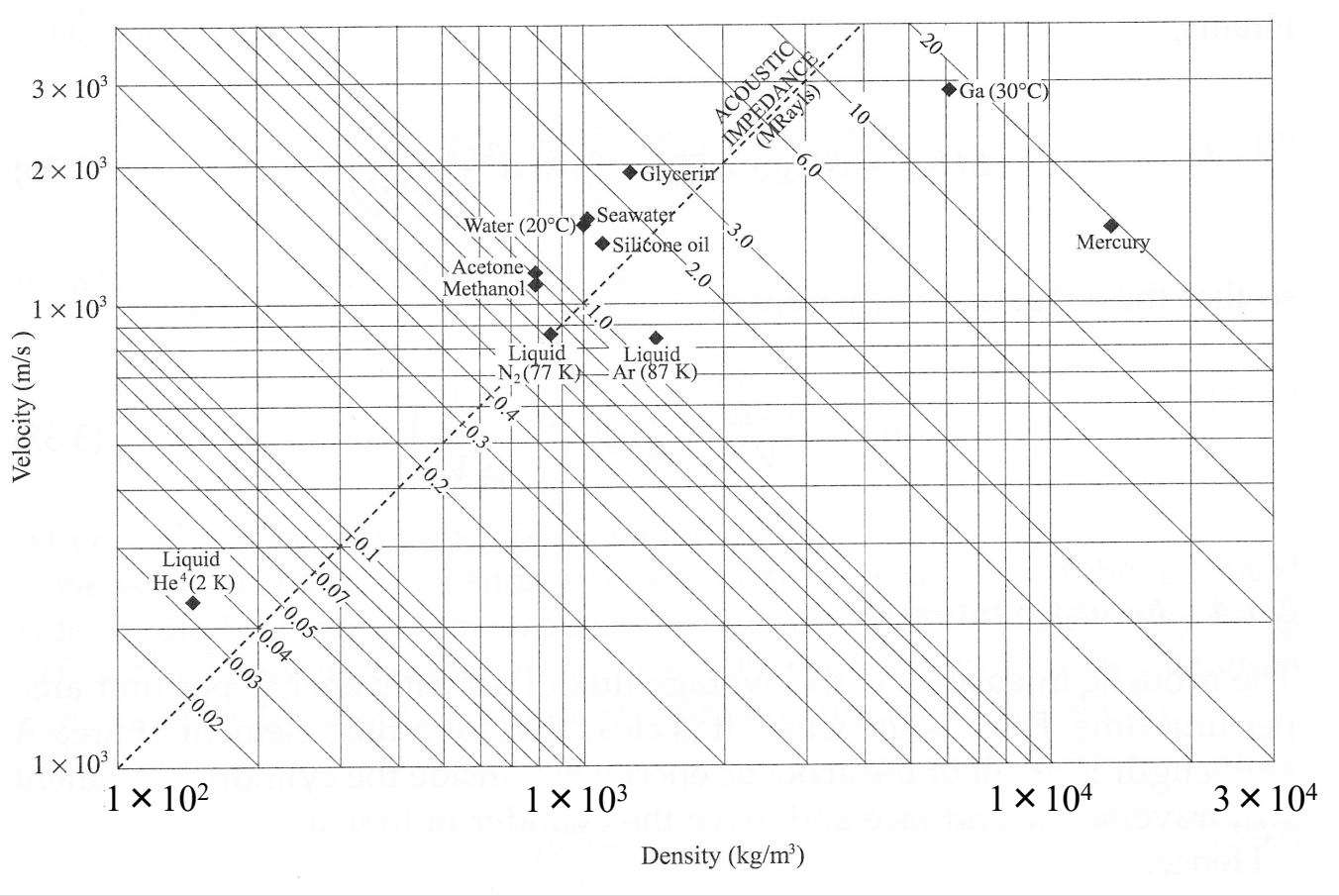
Unit: [Rayls],[Ns/m^3]

Acoustic characteristic impedance is determined by medium.

Typical acoustic characteristic impedance (fluid)

medium	ρ [kg/m³]	c [m/s]	ρc [Pa·s/m]
Water(20℃)	998	1481	1.48×10^{6}
Ethyl alcohol (20℃)	1026	1500	1.54×10^{6}
Air(20℃, 1atm)	1.2	343	4.15×10^{2}
Hydrogen(20℃, 1atm)	0.09	1308	1.18×10^{2}
Helium(0℃, 1atm)	0.179	970	1.73×10^{2}

Relation between density and sound velocity



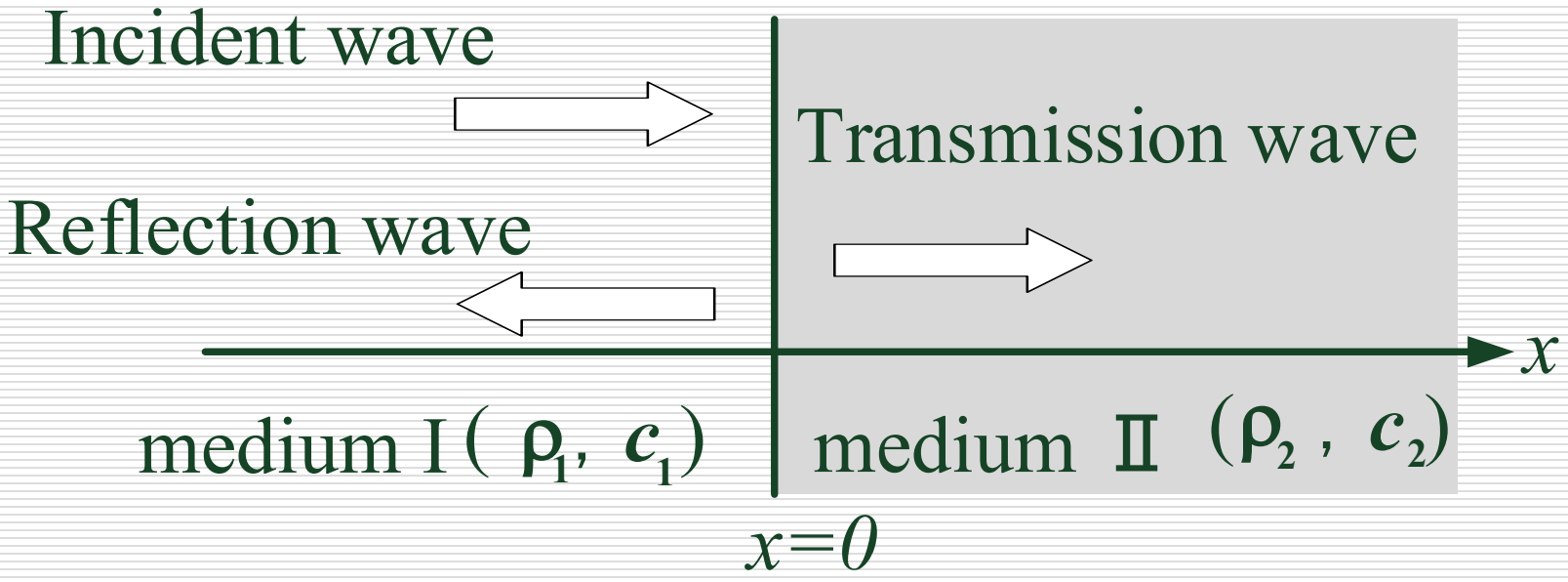
※J.David N.Cheeke: Fundamentals and Applications of Ultrasonic Waves, CRC Press (2002)

Typical acoustic characteristic impedance (solid)

medium	ρ [kg/m³]	c [m/s]	ρc [Pa·s/m]
Steel	7800	5950	27.4×10^6
Gold	19320	3240	62.5×10^6
Glass	2320	5640	13.1×10^6
Quartz	2650	5720	15.16×10^6
Diamond	3520	18000	63.4×10^6
Paper	800	2200	1.76×10^6

Transmission and reflection

Reflection wave and transmission wave occur when propagated wave incidents to the boundary where the acoustic characteristic impedance changes.



Equations at the boundary

Amplitude of incident, reflection and transmission wave indicates as C_1, C_2 and C_3, respectively.

$$\Phi(x,t) = C_1 e^{-jkx} + C_2 e^{jkx} \quad (x < 0)$$

$$\Phi(x,t) = C_3 e^{-jkx} \quad (x > 0)$$

Since sound pressure and particle velocity must be continued at the boundary, the following equations are obtained.

$$\rho_1 (C_1 + C_2) = \rho_2 C_3 \quad , \quad k_1 (C_1 - C_2) = k_2 C_3$$

Sound pressure reflection coefficient, and sound pressure transmission coefficient

Sound pressure reflection coefficient *R*

$$\frac{\omega\rho_1 C_2}{\omega\rho_1 C_1} = \frac{\rho_2 c_2 - \rho_1 c_1}{\rho_2 c_2 + \rho_1 c_1} \quad (= R)$$

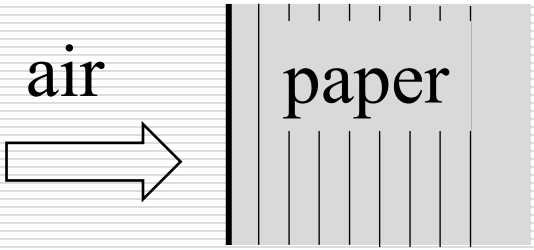
Sound pressure transmission coefficient *T*

$$\frac{\omega\rho_2 C_3}{\omega\rho_1 C_1} = \frac{2\rho_2 c_2}{\rho_2 c_2 + \rho_1 c_1} \quad (= T)$$

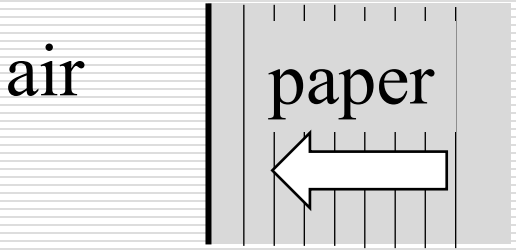
(Previous) Question 3

Calculate the reflection coefficient in the following cases.

1） A sound wave is incident on a stack of paper from the air.



2） A sound wave is incident on air from a stack of paper.



(Answer) Question 3

1) Air to paper

Reflection coefficient $R = \frac{1.76 \times 10^6 - 4.15 \times 10^2}{1.76 \times 10^6 + 4.15 \times 10^2} \approx 1$

2) Paper to air

Reflection coefficient $R = \frac{4.15 \times 10^2 - 1.76 \times 10^6}{4.15 \times 10^2 + 1.76 \times 10^6} \approx -1$

$|R| = 1$ indicates complete reflection because the acoustic impedance of air is much smaller than that of paper. The sign represents the phase of the reflected wave relative to the incident wave. A negative sign indicates a 180-degree phase shift.

Sound propagation in three dimensions

Operation of the sound in the three dimensions

In the three dimensions, the particle velocity has to be treated as vector.

$$\boldsymbol{v} = (v_x , v_y , v_z)$$

Also, partial derive of the spatial variable has to be operated in three dimensions as following

$$\frac{\partial}{\partial x} \quad \Rightarrow \quad \nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

Wave equation in the three dimensions

Equation of motion : $$\rho_0 \frac{\partial \mathbf{v}}{\partial t} = -\nabla p = \left(-\frac{\partial p}{\partial x}, -\frac{\partial p}{\partial y}, -\frac{\partial p}{\partial z} \right)$$

Conservation of mass: $$\frac{\partial \rho}{\partial t} = -\rho_0 \nabla \bullet \mathbf{v} = -\rho_0 \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

Gas state equation: $$p = c^2 \rho \quad (where \quad c^2 = \frac{\gamma P_0}{\rho_0})$$

Wave equation in the three dimensions

$$\nabla^2 p = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2}$$

Helmholtz equation

In the acoustic phenomena, the time variation is expressed by the harmonic function $e^{j\omega t}$

Sound pressure

$$p = \rho \frac{\partial \Phi}{\partial t} = j\omega\rho\Phi$$

Particle velocity

$$\mathbf{v} = -\nabla\Phi$$

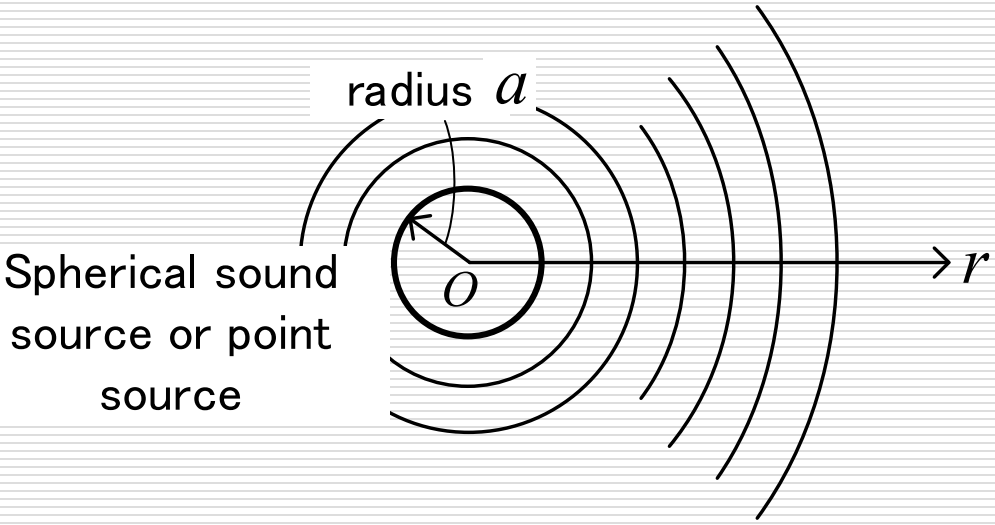
By substituting above equation to the wave equation, the following equation "Helmholtz equation" is obtained.

$$\nabla^2\Phi + k^2\Phi = 0$$

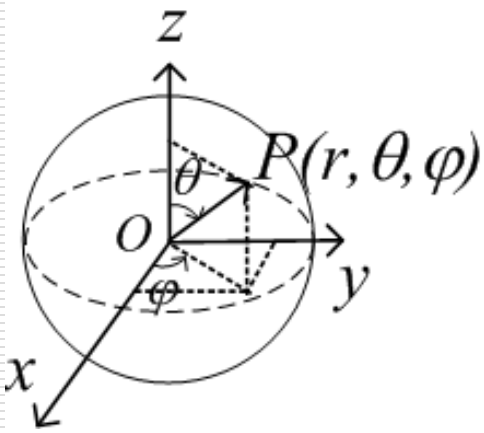
Polar coordinate system and point source

Propagation wave emitted from spherical sound source or point source is spherical wave which propagates uniformly in all directions.

Sound pressure is not distributed on the circumferential direction (θ, φ).



Polar coordinate system



Spherical wave

Wave equation in the polar coordinate system is following.

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \Phi}{\partial r} \right) + k^2 \Phi = 0$$

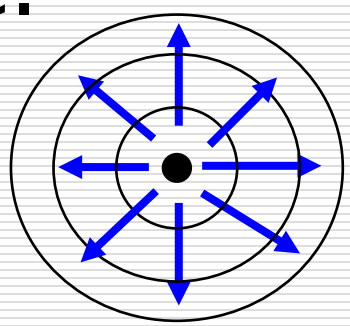
Solution of the above equation is following.

$$\Phi(x,t) = \left(\frac{C_1}{r} e^{-jkr} + \frac{C_2}{r} e^{jkr} \right) e^{j\omega t}$$

Froward and backward spherical wave

Forward wave is expansion into sound field from point source.

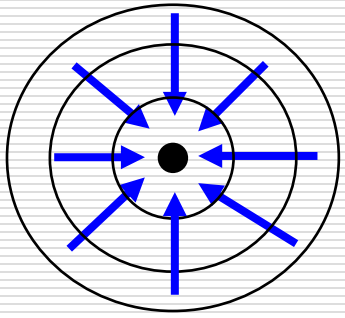
$$\Phi(x,t) = \frac{C_1}{r} e^{j(\omega t - kr)}$$



We only have to consider forward wave in 3 dimensional propagation.

Backward wave is contraction into point source. Generally, it's hard to propagate.

$$\Phi(x,t) = \frac{C_2}{r} e^{j(\omega t + kr)}$$



Sound wave from point source: sound pressure p

In the case of no reflection that is, only forward wave exists, velocity potential is as following,

$$\Phi(r,t) = \frac{C_1}{r} e^{j(\omega t - kr)}$$

Generally, only forward wave propagates from the spherical point source.

Sound pressure p can be can be obtained by the time derivative of the velocity potential as following,

$$p(r,t) = \rho \frac{\partial \Phi}{\partial t} = j\omega\rho\Phi$$

Hereafter, static density is expressed by ρ instated of ρ_0.

Sound wave from point source: particle velocity v

Particle velocity v can be obtained by the spatial derivative of the velocity potential as following,

$$v = -\frac{\partial \Phi}{\partial r} = -\frac{\partial}{\partial r}\left(\frac{C_1}{r} e^{j(\omega t - jkr)}\right) = -\frac{\partial}{\partial r}\left(\frac{C_1}{r}\right) e^{j(\omega t - jkr)} - \frac{C_1}{r}\frac{\partial\left(e^{j(\omega t - jkr)}\right)}{\partial r}$$

$$= -\left(-\frac{C_1}{r^2} e^{j(\omega t - jkr)}\right) - \frac{C_1}{r}(-jkr) e^{j(\omega t - jkr)} = \frac{C_1}{r^2}(1 + jkr) e^{j(\omega t - jkr)}$$

$$= \frac{1 + jkr}{r}\Phi$$